p8106_hw2_sc5826

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Data Cleaning & Splitting into Train/Test

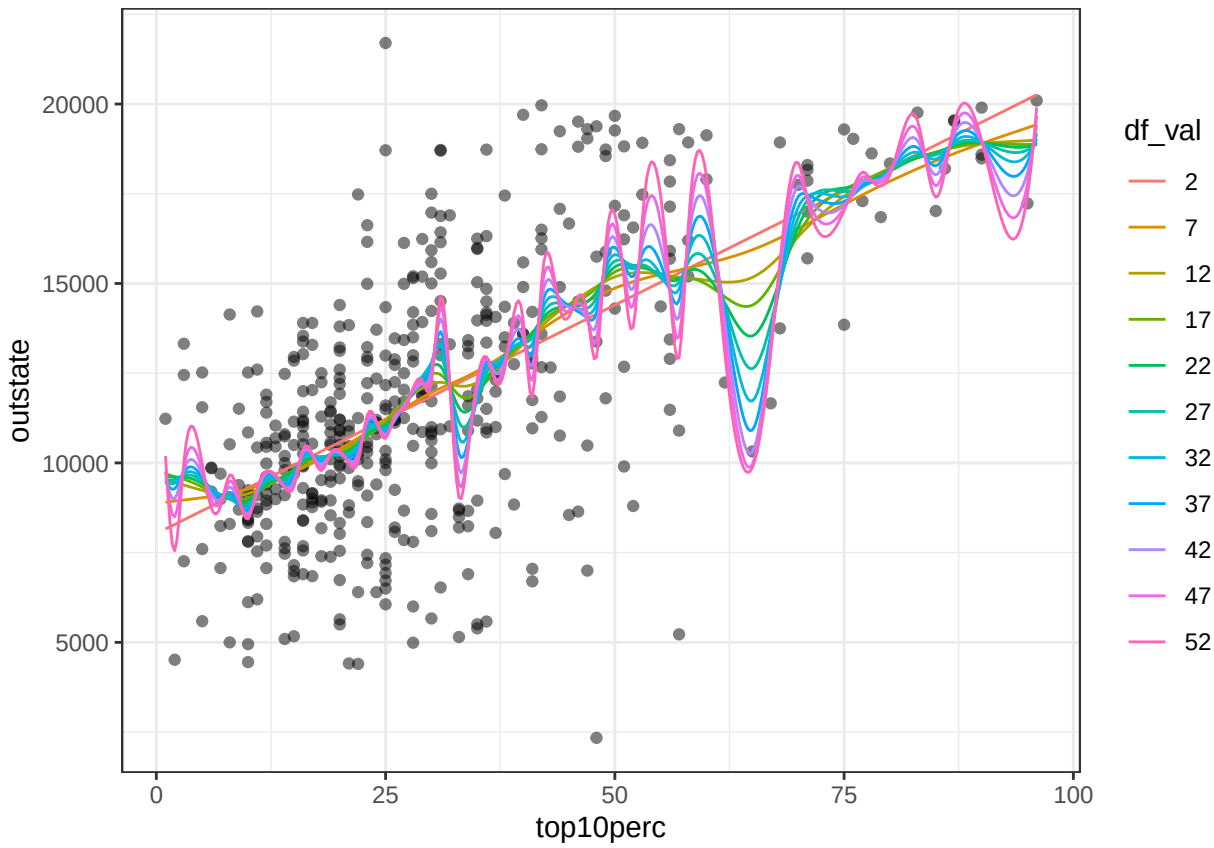
```
college = read_csv("College.csv") |>
  janitor::clean_names()
```

```
## Rows: 565 Columns: 18
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (1): College
## dbl (17): Apps, Accept, Enroll, Top10perc, Top25perc, F.Undergrad, P.Undergr...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
set.seed(1234)
indexTrain = createDataPartition(y = college$outstate, p = 0.8, list = FALSE)
college_train = college[indexTrain, ]
college_test = college[-indexTrain, ]
```

A - Smoothing Spline

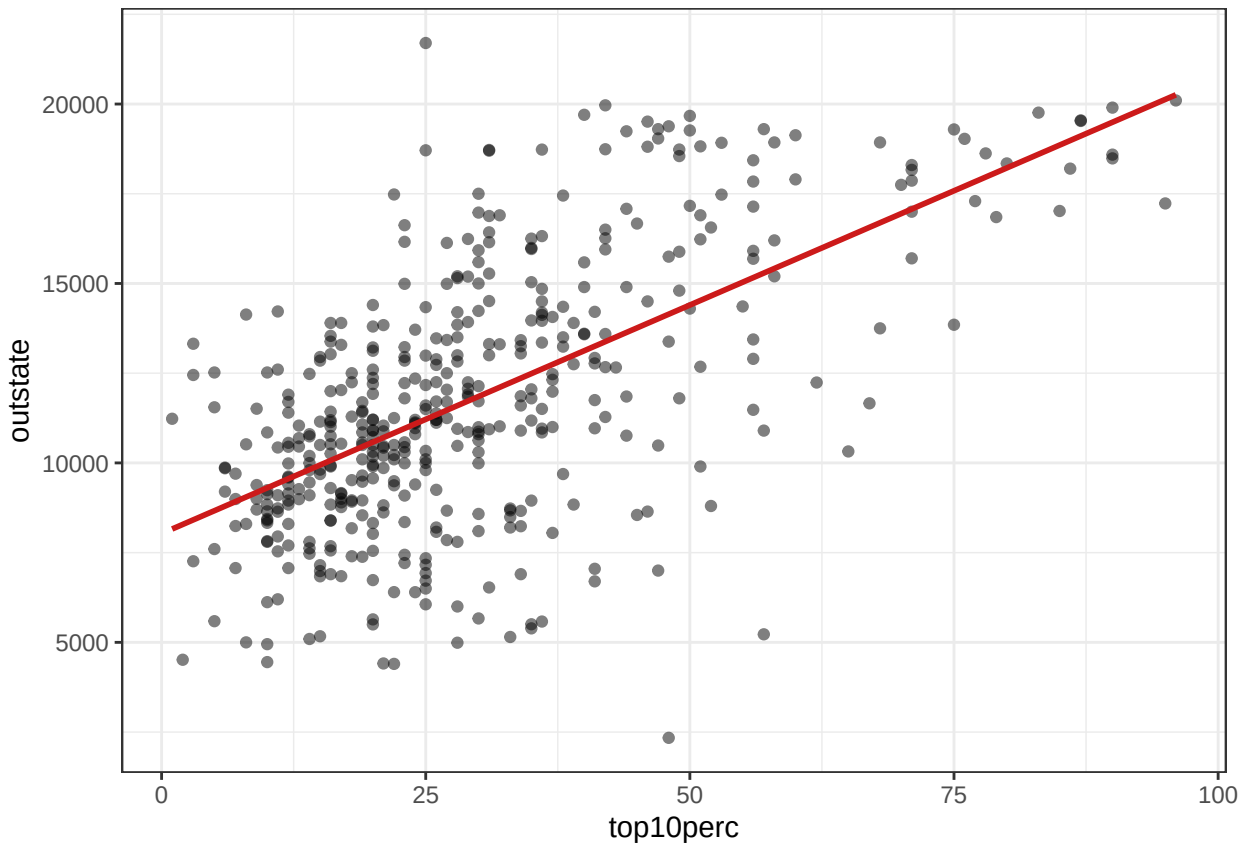
```
set.seed(1234)
top10perc_grid = seq(min(college_train$top10perc), max(college_train$top10perc), by = 0.25)

spline_df_fits =
  tibble(df_val = seq(from = 2, to = 52, by = 5)) |>
  mutate(
    spline_model = map(df_val, \(df) smooth.spline(x = college_train$top10perc, y = college_train$outsta
    predictions = map(spline_model, \(model) predict(model, x = top10perc_grid) |> as.data.frame())
  ) |>
  unnest(predictions) |>
  mutate(df_val = as.factor(df_val))

ggplot(college_train, aes(x = top10perc, y = outstate)) +
  geom_point(alpha = 0.5) +
  geom_line(data = spline_df_fits, aes(x = x, y = y, color = df_val)) +
  theme_bw()
```



As the degrees of freedom increase, the smoothing spline line deviates more from a straight line. At the lowest degrees of freedom (`df = 2`), the model has a heavier penalty for roughness, resulting in a nearly straight line. At the other end (`df = 52`), there is a lighter penalty for roughness, and the spline adjusts more to individual points. At higher degrees of freedom, there is a risk of overfitting the training data.

To select the optimal degrees of freedom, I used the default smoothing spline function, without specifying the degrees of freedome. This calculates and minimizes the default Generalized Cross-Validation (GCV) error to select the tuning parameter (λ).

```
best_smoothing_spline = smooth.spline(x = college_train$top10perc, y = college_train$outstate)

optimal_df = best_smoothing_spline$df

optimal_pred_df = predict(best_smoothing_spline, x = top10perc_grid) |>
  as.data.frame()

ggplot(college_train, aes(x = top10perc, y = outstate)) +
  geom_point(alpha = 0.5) +
  geom_line(data = optimal_pred_df, aes(x = x, y = y), color = rgb(.8, .1, .1, 1), linewidth = 1) +
  theme_bw()
```

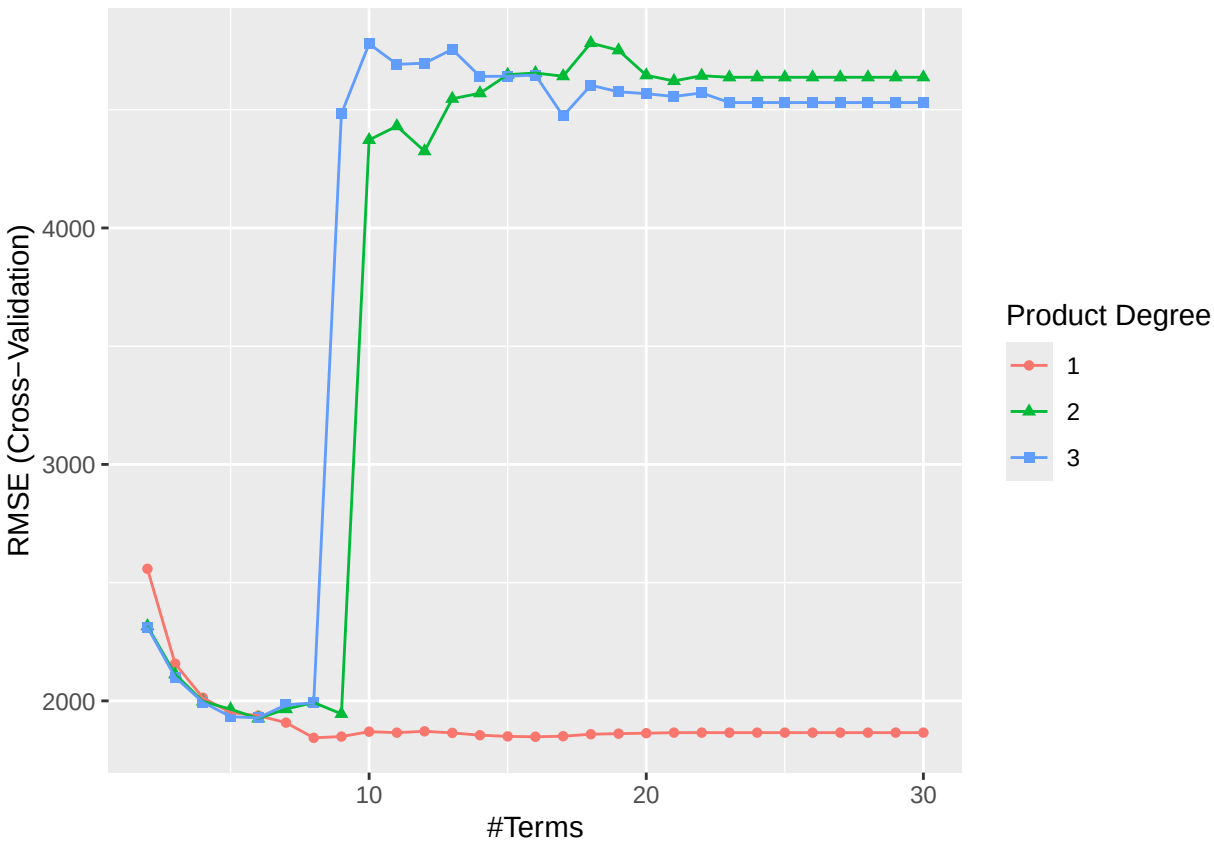


Based on the above criterion, the optimal degrees of freedom chosen for the smoothing spline model was ~2.

B - MARS

Train a multivariate adaptive regression spline (MARS) model to predict the response variable. Report the regression function. Present the partial dependence plot of an arbitrary predictor in your model. Report the test error.

```
# matrix of predictors
x = model.matrix(outstate ~ . - college, college_train)[, -1]
# vector of response
y = college_train$outstate

mars_grid = expand.grid(degree = 1:3, nprune = 2:30)

set.seed(1234)

ctrl1 = trainControl(method = "cv",
                     number = 10)

mars_model = train(x, y,
                   method = "earth",
                   tuneGrid = mars_grid,
                   trControl = ctrl1)

ggplot(mars_model)
```




```r
mars_model$bestTune
```

```
##   nprune degree
## 7      8      1
```

Regression Function:

```r
summary(mars_model$finalModel)
```

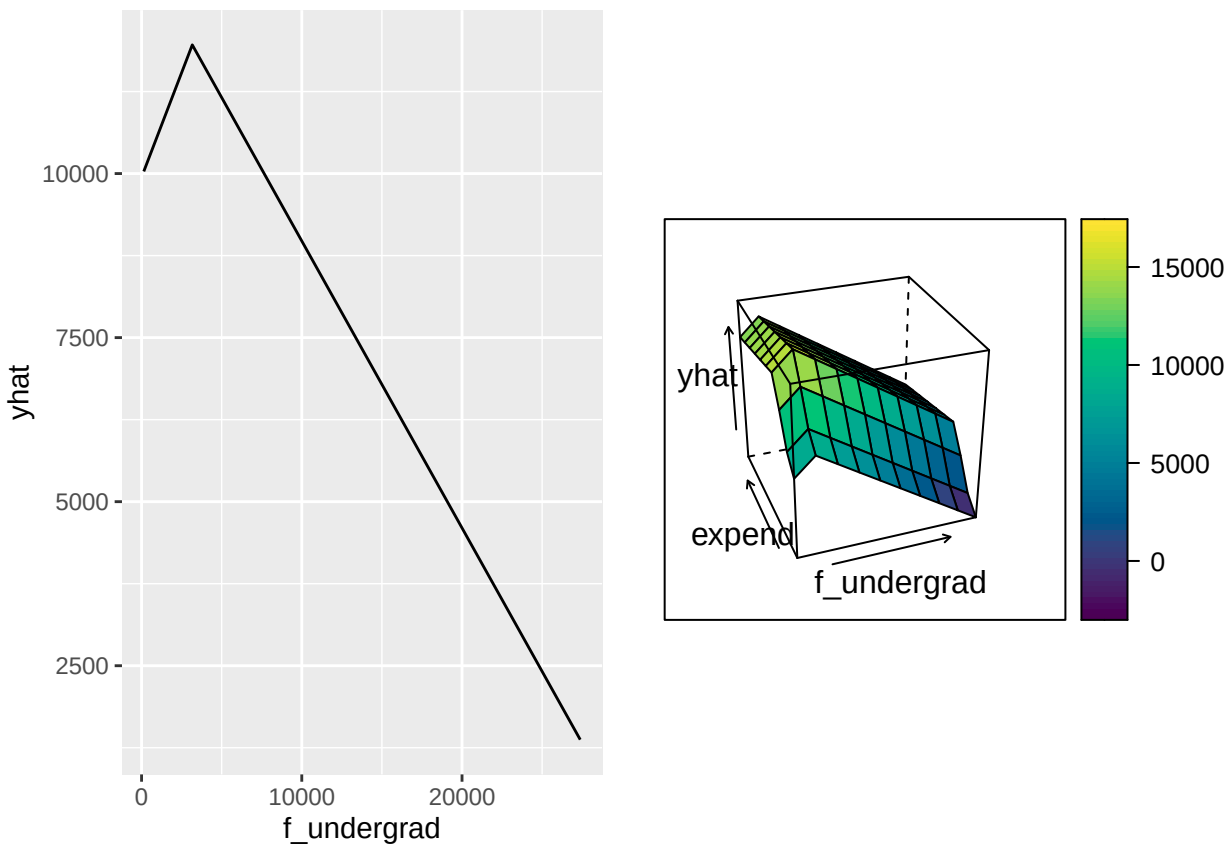
```
## Call: earth(x=matrix[453,16], y=c(7440,11250,75...), keepxy=TRUE, degree=1,
##             nprune=8)
##
##                     coefficients
## (Intercept)            9221.1968
## h(apps-2306)              0.4164
## h(1379-f_undergrad)      -2.1852
## h(f_undergrad-1379)      -0.4373
## h(4250-room_board)       -1.9481
## h(perc_alumni-14)        51.9742
## h(expend-5391)            0.7316
## h(expend-15494)          -0.8081
##
## Selected 8 of 28 terms, and 5 of 16 predictors (nprune=8)
## Termination condition: Reached nk 33
## Importance: expend, room_board, f_undergrad, perc_alumni, apps, ...
## Number of terms at each degree of interaction: 1 7 (additive model)
## GCV 3232127    RSS 1368795004    GRSq 0.765237    RSq 0.7795547
```

Final Regression Function:

$$
\begin{aligned}
\widehat{Outstate} = 9221.1968 \\
&+ 0.4164 * \text{h(apps-2306)} \\
&- 2.1852 * \text{h(1379-f.undergrad)} \\
&- 0.4373 * \text{h(f.undergrad-1379)} \\
&- 1.9481 * \text{h(4250-room.board)} \\
&+ 51.9742 * \text{h(perc.alumni-14)} \\
&+ 0.7316 * \text{h(expend-5391)} \\
&- 0.8081 * \text{h(expend-15494)}
\end{aligned}
$$

Partial Dependence Plot:

```
pdp_plot_1 = pdp::partial(mars_model, pred.var = c("f_undergrad"),
                          grid.resolution = 10) |>
  autoplot()

pdp_plot_2 = pdp::partial(mars_model, pred.var = c("f_undergrad", "expend"),
                          grid.resolution = 10) |>
  pdp::plotPartial(levelplot = FALSE, zlab = "yhat", drape = TRUE, screen = list(z = 20, x = -60))

gridExtra::grid.arrange(pdp_plot_1, pdp_plot_2, ncol = 2)
```



Test Error:

```
predictions = predict(mars_model, newdata = college_test)

test_rmse = RMSE(predictions, college_test$outstate)
```

The test error (RMSE) for the MARS model is 1861.76.